Executive Function and Cognitive Behavioral Therapy for Adult ADHD: A Meta-Review Protocol and Critical Introduction

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October 21, 2026
Protocol version 0.1; first manuscript draft

Abstract

Cognitive behavioral therapy (CBT) has evidence of short-term benefit for adult attention-deficit/hyperactivity disorder (ADHD), but the certainty and interpretation of that evidence depend on comparator and study design (Lopez et al., 2018). The motivating question here is whether baseline executive function identifies who benefits, rather than whether executive function can improve during treatment. Those are different estimands: prediction concerns later outcome, whereas moderation concerns variation in a treatment contrast (Kraemer et al., 2002). We propose a meta-review of systematic reviews, supplemented by original adult-ADHD studies measuring baseline executive function and subsequent CBT outcomes. The protocol separates cognitive restructuring from multicomponent packages, prognostic associations from treatment interactions, and active from passive comparators. It specifies ten search slices across Europe PMC, PubMed and PsycINFO, independent screening, review-overlap mapping and structured extraction of estimates and uncertainty. This initial report provides an executed Europe PMC count baseline, an auditable citation ledger and preserved author drafts; it does not report completed screening or a pooled treatment effect. The executive-demand critique remains a hypothesis requiring direct tests, alongside evidence for benefit, null findings and alternative explanations.

1 Introduction

Evidence for cognitive behavioral therapy (CBT) in adult attention-deficit/hyperactivity disorder (ADHD) warrants a question more precise than whether the treatment "works." A Cochrane review identified possible short-term benefits but emphasized limitations in evidence quality and the need for longer-term evidence (Lopez et al., 2018). A randomized trial in medication-treated adults with persistent symptoms found greater improvement with CBT than with relaxation and educational support (Safren et al., 2010). That result concerns a particular population and treatment comparison; the present review asks whether baseline executive function modifies such treatment contrasts. Recent meta-analyses also examine executive-function outcomes and everyday functioning, providing reasons to distinguish what improves from who benefits (Liu et al., 2026; Lopez-Pinar et al., 2026).

The author's archived blog and prior critical review motivate a narrower hypothesis: cognitive restructuring may be harder to use when executive resources are limited. We treat this as a proposed explanation to test, not as evidence that CBT is ineffective. Related pilot studies in older adults with anxiety or depression examine executive functioning and cognitive flexibility, but their populations and analyses do not directly answer the adult-ADHD moderation question (Mohlman and Gorman,

2005; Johnco et al., 2014). In particular, Johnco et al. (2014) did not find that pretreatment flexibility predicted overall treatment outcome, although associations with aspects of restructuring skill were reported. This mixed pattern cautions against turning a plausible cognitive-demand account into a settled clinical conclusion.

The distinction between a predictor, moderator and outcome is central. A baseline characteristic may predict subsequent improvement without changing the effect of CBT relative to another treatment; moderation concerns that difference in treatment effects (Kraemer et al., 2002). Similarly, evidence that executive function improves following CBT does not by itself test baseline executive function as a moderator; the outcome and interaction questions require different analyses (Liu et al., 2026; Kraemer et al., 2002). We will therefore extract the timing of executive-function measurement, the treatment contrast and the actual statistical model, rather than treating every mention of executive function and outcome as a moderator result. We will also distinguish multicomponent CBT from isolated cognitive restructuring, so that a component critique is evaluated at the level for which evidence exists.

This project contributes a prospective protocol and a reproducible search baseline for a critical meta-review, not a completed efficacy verdict. The primary synthesis will map systematic reviews of adult-ADHD CBT; a supplementary synthesis will identify original studies that can address baseline executive-function prediction or moderation. Search and selection reporting will follow the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework and its search extension (Page et al., 2021; Rethlefsen et al., 2021). The review is designed to retain supportive, contradictory and null evidence and to separate absence of a direct test from evidence of no moderation.

2 Methods and protocol

Protocol version 0.1 is dated October 21, 2026 and has not been registered. The full operational protocol is maintained in `protocol.md`. Eligible populations are adults with diagnosed ADHD; mixed samples require separately extractable adult-ADHD findings. Interventions are structured CBT programs, with cognitive restructuring, organization, planning, therapist support and delivery format extracted separately. Comparators are classified as passive, active psychological, or medication-only for adjunctive treatment. Primary clinical outcomes are ADHD symptoms and daily functioning, separated by rater and time point.

The meta-review includes systematic reviews with explicit search and selection methods. The supplementary original-study stream requires pretreatment executive-function measurement and a later outcome association or treatment interaction. Single-arm associations are prognostic evidence, not causal treatment-effect moderation. Post-treatment executive-function scores alone do not enter the baseline moderator synthesis. We will record whether measures are performance-based or self-reported, as well as diagnosis, medication, comorbidity, outcome overlap, missingness and prespecification.

Reviews will be appraised using AMSTAR 2, with domain judgments retained (Shea et al., 2017). The original-study risk-of-bias instrument will be fixed in a protocol amendment before extraction. Review overlap will be mapped at the primary-study level; overlapping meta-analyses will not be pooled as independent estimates. Quantitative moderation synthesis remains conditional on compatible constructs, comparisons and interaction estimands, with a statistical analysis amendment required before fitting.

3 Search strategy

Ten overlapping search slices cover reviews, trials, executive-function associations, broader executive-function terms, specific cognitive processes, medication, components and comparators, adherence, functioning and anxiety. The complete Europe PMC, PubMed and PsycINFO translations are versioned in `search/queries.json`; Europe PMC strings are reproduced in the protocol. The baseline publication-date cutoff is October 21, 2026. Only Europe PMC has been executed in this initialization. PubMed and PsycINFO remain planned; the Ovid PsycINFO translations require institutional syntax validation.

The standard-library Python script retrieves match counts, preserves request queries and response hashes, and appends dated rows to a comma-separated values (CSV) log. Each response includes only one record identifier, so this operation is not a complete screening export. A failed request is logged with a blank count and an error, never as zero. The historical notes of 27 reviews and 9 executive-function/moderator/outcome hits lack their exact search strings and dates and remain unverified. Current queries are not tuned to reproduce those numbers. Complete record exports, deduplication, reference checking and search sensitivity checks are required before eligibility or completeness claims, consistent with search-reporting guidance (Rethlefsen et al., 2021).

4 Screening and extraction plan

Two independent reviewers will pilot the eligibility rules and screen titles/abstracts and full texts, with an adjudicator resolving remaining disagreements. Reviewers have not yet been assigned, and no independent screening is claimed in this draft. Full-text exclusions will receive explicit reasons; unavailable texts will remain awaiting retrieval. Companion reports will be linked to underlying studies, and duplicates across databases and reviews will be tracked.

Extraction will record outcome-specific denominators, intervention components, executive-function timing and instrument, comparison group, rater, follow-up, estimate, confidence interval, reported significance test, covariates, missingness and prespecification. One-arm associations, between-arm interactions and changes in executive function will be labeled separately. The selection flow will be populated only after screening, following PRISMA reporting guidance (Page et al., 2021).

5 Results: initialization and search baseline

All ten Europe PMC searches completed successfully on October 21, 2026, with the publication-date cutoff fixed to that day. Table 1 reports the endpoint match counts; these are not screened studies and overlap across slices. Counts are database retrieval totals, not inferential estimates, so confidence intervals and significance tests do not apply. No eligible-study total, pooled clinical estimate or moderator conclusion is available because screening has not been performed.

The source archive contains the author's blog and prior 29-page critical review. The current manuscript has ten independently identified bibliographic records, each linked to a scope-limited claim ledger. These initialization outputs establish provenance and a review plan, not a completed clinical evidence synthesis.

Slice	Search focus	Matches
S01	Reviews and meta-analyses	40
S02	Randomized trials	73
S03	Executive function and predictors or moderators	21
S04	Executive function sensitivity search	24
S05	Specific executive processes	10
S06	Medication and adjunctive treatment	63
S07	Active comparators and treatment components	22
S08	Adherence and feasibility	23
S09	Functioning and follow-up	71
S10	Comorbid anxiety	54

Table 1: Europe PMC baseline, protocol 0.1, run `baseline-10-21-2026`. All 10 requests succeeded on their first attempt. The complete queries and receipts are retained in the repository.

6 Discussion

The critical question is whether there is direct evidence that executive function changes the relative benefit of CBT, and whether that evidence concerns cognitive restructuring or a broader intervention. An average benefit in a trial and a proposed difficulty using one component are not mutually exclusive; the review will test how directly the available studies connect these claims. The cited clinical literature includes evidence favorable to CBT as well as limitations in certainty and design (Safren et al., 2010; Lopez et al., 2018; Lopez-Pinar et al., 2026). The author-source archive is preserved to make the motivation and prior argument inspectable, while the current protocol permits findings that challenge that argument.

The initial search is limited to database match counts and does not establish an eligible evidence set. Title/abstract terminology, database coverage, overlapping reviews, outcome choice and missing moderator analyses may constrain the eventual conclusions. Any inability to find a direct interaction test will be reported as an evidence gap, not as proof that CBT cannot help people with executive-function difficulties. No individual treatment recommendation follows from this protocol.

References

C Johnco, V M Wuthrich, and R M Rapee. The influence of cognitive flexibility on treatment outcome and cognitive restructuring skill acquisition during cognitive behavioural treatment for anxiety and depression in older adults: Results of a pilot study. *Behaviour research and therapy*, 57:55–64, 2014. doi: 10.1016/j.brat.2014.04.005. URL `https://pubmed.ncbi.nlm.nih.gov/24828838/`.

Helena Chmura Kraemer, G Terence Wilson, Christopher G Fairburn, and W Stewart Agras. Mediators and moderators of treatment effects in randomized clinical trials. *Archives of general psychiatry*, 59(10):877–883, 2002. doi: 10.1001/archpsyc.59.10.877. URL `https://pubmed.ncbi.nlm.nih.gov/12365874/`.

Yang Liu, Feilong Zhu, Yueyong Yu, Yiqi Song, Hao Luo, Baohua Xu, Shengqi Lin, Zhitong Ha, Dongqing Kuang, Yongtao Xie, Zhexue Chen, and Yuanchun Ren. A meta-analysis of the intervention effect of cognitive behavioral therapy on adult ADHD. *Journal of affective disorders*, 399:121107, 2026. doi: 10.1016/j.jad.2025.121107. URL `https://pubmed.ncbi.nlm.nih.gov/41483880/`.

Pablo Luis Lopez, Fernando Manuel Torrente, Agustin Ciapponi, Alicia Graciela Lischinsky, Marcelo Cetkovich-Bakmas, Juan Ignacio Rojas, Marina Romano, and Facundo F Manes. Cognitive-behavioural interventions for attention deficit hyperactivity disorder (ADHD) in adults. *The Cochrane database of systematic reviews*, 3:CD010840, 2018. doi: 10.1002/14651858.cd010840.pub2. URL `https://pubmed.ncbi.nlm.nih.gov/29566425/`.

Carlos Lopez-Pinar, Benjamin Selaskowski, Marcel Schulze, Martin Oscarsson, Javier Fenollar-Cortes, Margaret H Sibley, Alexandra Philipsen, and Tatja Hirvikoski. Cognitive behavioral therapy effects on global functioning, domain-specific functioning, and quality of life in adult ADHD: A comprehensive meta-analysis. *Behaviour research and therapy*, 201:105026, 2026. doi: 10.1016/j.brat.2026.105026. URL `https://pubmed.ncbi.nlm.nih.gov/41936228/`.

Jan Mohlman and Jack M Gorman. The role of executive functioning in CBT: a pilot study with anxious older adults. *Behaviour research and therapy*, 43(4):447–465, 2005. doi: 10.1016/j.brat.2004.03.007. URL `https://pubmed.ncbi.nlm.nih.gov/15701356/`.

Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, Roger Chou, Julie Glanville, Jeremy M Grimshaw, Asbjrn Hrobjartsson, Manoj M Lalu, Tianjing Li, Elizabeth W Loder, Evan Mayo-Wilson, Steve McDonald, Luke A McGuinness, Lesley A Stewart, James Thomas, Andrea C Tricco, Vivian A Welch, Penny Whiting, and David Moher. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ (Clinical research ed.)*, 372:n71, 2021. doi: 10.1136/bmj.n71. URL `https://pubmed.ncbi.nlm.nih.gov/33782057/`.

Melissa L Rethlefsen, Shona Kirtley, Siw Waffenschmidt, Ana Patricia Ayala, David Moher, Matthew J Page, Jonathan B Koffel, and PRISMA-S Group. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. *Systematic reviews*, 10(1):39, 2021. doi: 10.1186/s13643-020-01542-z. URL `https://pubmed.ncbi.nlm.nih.gov/33499930/`.

Steven A Safren, Susan Sprich, Matthew J Mimiaga, Craig Surman, Laura Knouse, Meghan Groves, and Michael W Otto. Cognitive behavioral therapy vs relaxation with educational support for medication-treated adults with ADHD and persistent symptoms: a randomized controlled trial. *JAMA*, 304(8):875–880, 2010. doi: 10.1001/jama.2010.1192. URL `https://pubmed.ncbi.nlm.nih.gov/20736471/`.

Beverley J Shea, Barnaby C Reeves, George Wells, Micere Thuku, Candyce Hamel, Julian Moran, David Moher, Peter Tugwell, Vivian Welch, Elizabeth Kristjansson, and David A Henry. AMSTAR 2: a critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. *BMJ (Clinical research ed.)*, 358:j4008, 2017. doi: 10.1136/bmj.j4008. URL `https://pubmed.ncbi.nlm.nih.gov/28935701/`.

A Source archive and reproducibility

The repository preserves the author's original blog Markdown, a Google document text export, and the 29-page deep-research Draft 2 PDF with extracted text. Source identifiers and SHA-256 hashes are recorded in `sources/manifest.json`. The imported review covers a broader anxiety and cognitive-restructuring question and identifies itself as a critical theoretical review rather than a completed systematic review. Its claim of approximately 300 identifier-verified sources is not

independently certified here; the underlying ledger was not retrieved. Its journal-style submission footer is not evidence of an actual submission.

The current bibliography uses separately retrieved Europe PMC metadata, with a claim-and-scope ledger in `data/citation-evidence.csv`. Metadata and abstract checks establish source identity and support the limited statements used here; they do not replace full-text assessment for the review. Run `make test` for deterministic checks and `make paper` to compile this manuscript with latexmk and BibTeX. Search reruns use a new run label and retain earlier receipts. The public repository is `https://github.com/brando90/meta-review-cbt`.